

Table 5: ROUGE-L of summaries generated by GPT-4 on the SQuALITY validation set under two settings: baseline and incremental updating.

Chunk size	ROUGE-L
<i>Baseline</i>	
-	14.9
<i>Incremental updating</i>	
256	13.2
512	13.7

E.1 QUALITATIVE ANALYSIS

We show the baseline, incremental, and human summaries for *Venus Is a Man’s World* from SQuALITY in Table 6. The incremental summary was generated by GPT-4 with a chunk size of 512.

The baseline summary provides the most coherent and comprehensive overview of the story’s key events and themes. It clearly introduces the main characters - Ford, Evelyn, and Butt - and their backgrounds, while succinctly summarizing the plot including Ford’s friendship with Butt, Evelyn’s discovery and confrontation, and Butt’s unconventional marriage proposal. The summary highlights central conflicts related to gender roles and norms. In contrast, the incremental summary, while touching on similar plot points and themes, does so in a more disjointed, less cohesive manner. Details about the setting, dunging drug harvesting, and weaponry feel redundant. Finally, the human summary lacks clarity in many parts, with confusing references to the Male Desuffrage Act and inconsistent character details. It focuses heavily on the early parts of the plot at the expense of later key events.

Overall, our finding that GPT-4 summaries can be better than human summaries aligns with results from recent work on LLM-based short-document summarization (Pu et al., 2023b). The qualitative analysis shows that methods like incremental updating could introduce coherence errors into LLM-generated summaries.

F EFFECT OF SUMMARY LENGTH ON BOOOOKSCORE

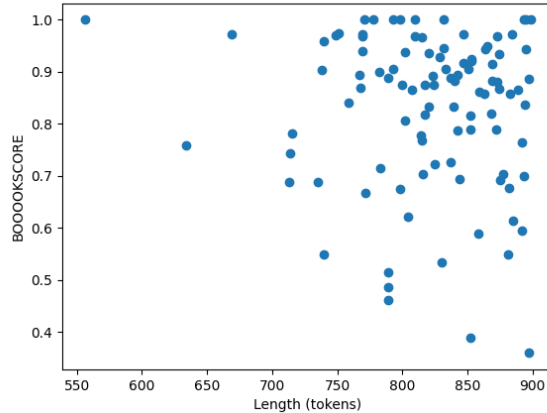


Figure 3: BOOOOKSCORE vs. summary length.

Table 6: Summaries of William Tenn’s *Venus Is a Man’s World*.

Baseline	Incremental updating	Human
Venus Is a Man’s World is a story about a young boy named Ferdinand, or “Ford,” and his sister Evelyn, who are traveling on a spaceship called the Eleanor Roosevelt to the planet Venus. Evelyn is seeking a husband on Venus, as there is a shortage of men on Earth due to wars and other factors. Ford, curious and adventurous, explores the ship and stumbles upon a stowaway named Butt Lee Brown, a Venusian farmer who had gotten into trouble on Earth and is now hiding in a lifeboat. Ford and Butt form a bond, with Butt teaching Ford about Venusian culture, songs, and even politics. Ford starts to admire Butt’s strong, independent nature and begins to question the Earth’s female-dominated society. However, when Evelyn discovers Ford’s friendship with Butt, she confronts the stowaway, accusing him of corrupting her brother’s morals and breaking the law by stowing away on the ship. Butt, unfazed by Evelyn’s accusations, proposes marriage to her, stating that they are both on the same journey to find a partner and that she has the qualities he is looking for in a wife. Ford excitedly urges his sister to accept Butt’s proposal, but the story leaves the outcome of this unconventional proposal unresolved. The story highlights the clash of gender roles, societal norms, and the challenges faced by individuals seeking love and companionship in a rapidly changing world.	In the story, siblings Ford and Evelyn from an underwater city accompany women seeking husbands on a journey to Venus. Ford befriends stowaway, Butt Lee Brown, who has a troubled past. They bond over their life stories as Butt educates Ford about Venusian culture and weaponry. Despite being outsiders, they travel together. Evelyn convinces Ford to attend a geography lecture, where she becomes deeply engaged in the discussion of illegal dunging drug harvesting on the Macro Continent. Ford’s knowledge of dunging and the Venusian term “Paddlefoot” piques Evelyn’s interest. Ford argues that Venusians are pioneers who need strong, independent men for building a civilization on the new planet, thus intriguing Evelyn. Ford takes Evelyn to meet Butt Lee Brown, making her promise not to turn him in. Upon meeting, Evelyn confronts Butt about his illegal activities and moral delinquency. Butt dismisses her concerns, sparking a debate between them about Earth and Venusian societies. During their conversation, Butt proposes to Evelyn, stating that they both have similar goals in finding a spouse and that she possesses qualities he finds desirable. Evelyn is taken aback by the proposal, and her brother Ford excitedly encourages her to accept. The story continues to explore the characters’ relationships and their differing perspectives on the new planet, as well as the potential consequences of their actions and decisions.	Ferdinand is on a spaceship with his elder sister. The ship is jam-packed with females going to Venus in search of husband and he exits the cabin while the women are still in their hammocks. The women at his times have all the rights and do all the important things since the Male Desuffrage Act, so the boy is admitted to the ship on behalf of his sister. He explores the empty ship in search of portholes and after some hesitation enters a forbidden area. There he looks at the stars and then tries to open a lock on the lifeboat. A huge scary man appears with a blaster and frighteningly cold gaze. Ferdinand explains that he comes from Undersea, an area on Earth, and tells his family story - his parents being one of the first married couples in Undersea and dying a while ago, leading to his sister’s decision to migrate to Venus. The stranger, Butt, tells about the lack of women on Venus and his travel to Earth in search of a wife without any idea “it’s a woman’s world”. So he got in trouble with the law and stowed away on this ship. His many brothers were killed in a rising and only one is left. From that day on Ferdinand keeps visiting the stowaway bringing fruits and listening to stories about Venus. Butt teaches the boy to use the blaster without giving it not hold and constantly asks about Evelyn, the sister. Once, Ferdinand attends a geography lecture on the ship with his sister and corrects the lecturer about Venusian geography. Evelyn starts eliciting where the boy learned that and the boy tells about real men working on Venus. Sis gets angry with those masculine ideas and doesn’t believe them to come from a little boy. Ferdinand tries to lie but Sis suppresses him into confession and he leads her to Butt. She tells Butt about all the laws he has broken while the least responds with an appeal to sense. Suddenly, Butt simply proposes a mutually beneficial marriage to stop the debate.

To explore the impact of summary length on BOOOOKSCORE, we plot BOOOOKSCORE vs. length in Figure 3 for the 100 summaries generated by GPT-4 with incremental updating using a chunk size of 2048. The plot indicates that there is no discernible correlation between length and BOOOOKSCORE.

G BOOTSTRAPPING ANALYSIS OF BOOOOKSCORE

To check the stability of the BOOOOKSCORE metric, we ran a bootstrapping experiment. Given the BOOOOKSCORE of 100 summaries generated by GPT-4 using incremental updating with a chunk size of 2048, we randomly sample 1000 times with replacement using a sample size of 100, then compute the mean of these 1000 samples. The standard deviation of these samples is 0.015, which suggests that BOOOOKSCORE is consistent and reliable across multiple random samples.

H LABEL-WISE ALIGNMENT BETWEEN GPT-4 AND HUMAN ANNOTATIONS

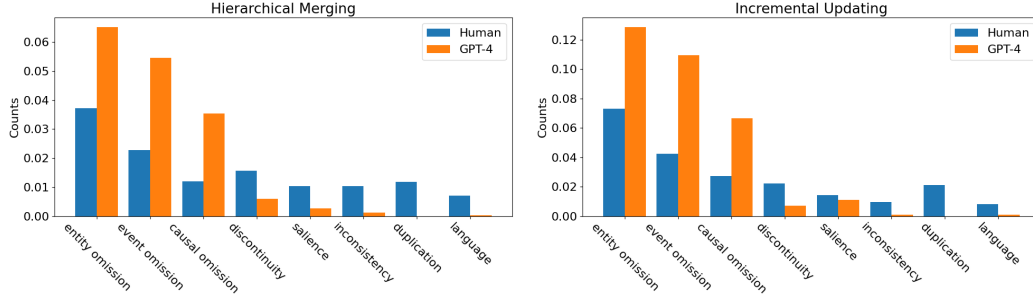


Figure 4: Distribution of all error types for incremental and hierarchical summaries generated by GPT-4, normalized by the total number of sentences in the summaries.

In our approach to GPT-4 automatic evaluation, we incorporate error type prediction into the prompt to assist the model in making reasoned judgments. We analyze the label-wise correlation between GPT-4 and human annotations, presenting the results in Figure 4. Despite the high agreement between precision as discussed in Section 4.1, GPT-4’s error distributions vary significantly from those of human annotations. GPT-4 demonstrates a strong propensity for labeling omission errors, occasionally applying these labels to sentences that could be more appropriately categorized under different error types. In this study, we have not delved deeper into the error types predicted by GPT-4 given the observed inaccuracy. Further analysis of GPT-4’s capability to precisely predict error types remains a potential area for future research.

I EFFECTIVENESS OF EXISTING REFERENCE-FREE EVALUATION METRICS

Table 7: Results from BLANC and SUPERT on all *hierarchical* summaries.

	BLANC	SUPERT
GPT-4 (4096)	0.0248	0.3627
GPT-4 (2048)	0.0221	0.3615
GPT-3.5-Turbo (2048)	0.0203	0.3658
Claude 2 (88000)	0.0171	0.349
Claude 2 (2048)	0.0167	0.3559
Mixtral-8x7B (2048)	0.0169	0.3836
LLaMA2-7B-Inst (2048)	0.013	0.3446

We investigate how well existing reference-free evaluation metrics work for our setting where the source text is at book level. BLANC (Vasilyev et al., 2020) measures how helpful a summary is to understanding the source document by testing if a pre-trained model can better fill in masked words given access to the summary. SUPERT (Gao et al., 2020) measures the semantic similarity between the a summary and some pseudo reference summaries generated from the source document. We compute both metrics for hierarchical summaries generated under all configurations, and show results in Table 7. Both BLANC and SUPERT differ by nearly negligible margins across models, which makes them less meaningful. Furthermore, both metrics rate Claude 2 (88000) summaries as inferior to GPT-3.5-Turbo (2048) summaries, a finding that clearly contradicts results from our qualitative analysis.